

DIMITROVGRAD, Serbia

WMO: 133970

Lat: 43.017N

Lon: 22.750E

Elev: 450

StdP: 96.03

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.7	-9.1	-14.6	1.1	-10.3	-11.9	1.4	-8.4	7.3	-1.1	6.5	-0.8	2.2	110	0.490

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.8	33.1	20.0	31.4	19.5	29.5	19.0	21.5	29.7	20.6	28.6	19.8	27.4	2.7	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.8	14.4	24.9	18.0	13.6	23.9	17.2	13.0	22.8	64.7	29.6	61.4	28.7	58.6	27.4	26.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.3	5.8	5.2	-16.3	35.7	3.7	1.9	-19.0	37.1	-21.2	38.2	-23.3	39.3	-26.0	40.7		
(5)			-16.5	23.0	3.6	1.2	-19.1	23.9	-21.2	24.6	-23.3	25.3	-25.9	26.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.9	-0.2	1.9	5.8	10.9	15.3	18.9	20.8	21.1	16.5	11.4	6.6	1.2		
	DBStd	8.43	4.69	4.96	4.46	3.95	3.41	3.28	2.82	2.93	3.65	4.24	4.47	4.55		
	HDD10.0	1154	317	228	141	35	3	0	0	0	1	37	119	273		
	HDD18.3	2978	576	460	387	225	102	32	9	9	76	218	353	531		
	CDD10.0	1481	0	1	12	61	169	266	336	343	197	78	17	1		
	CDD18.3	263	0	0	0	1	9	48	87	94	22	2	0	0		
	CDH23.3	3208	0	0	1	32	177	589	993	1097	289	28	1	0		
	CDH26.7	1179	0	0	0	5	31	193	392	466	89	3	0	0		
(14) Wind	WSAvg	2.6	2.8	2.8	2.9	2.7	2.5	2.3	2.3	2.3	2.3	2.4	2.7	2.8		
(15) Precipitation	PrecAvg	629	42	41	43	57	72	69	55	51	51	57	47	45		
	PrecMax	972	106	80	90	144	157	142	124	174	154	129	127	108		
	PrecMin	357	8	8	6	8	22	15	8	6	6	5	1	0		
	PrecStd	143	27	20	22	27	30	33	30	43	36	35	27	28		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.0	18.1	22.1	26.8	29.5	33.1	35.7	35.7	32.6	26.2	20.8	14.6		
		MCWB	8.2	9.9	11.9	15.1	17.7	20.6	20.9	19.9	18.3	16.2	13.7	10.3		
	2%	DB	9.9	14.2	19.1	23.4	27.1	30.8	33.0	33.3	29.2	23.4	17.7	11.7		
		MCWB	6.7	8.6	10.4	13.5	17.0	20.2	20.1	19.9	17.5	15.1	12.2	8.5		
	5%	DB	7.8	11.7	16.5	20.9	25.3	28.9	30.9	31.4	26.7	21.2	15.3	9.5		
		MCWB	5.4	7.3	9.3	12.2	16.2	19.5	19.6	19.3	16.7	14.4	11.0	7.2		
	10%	DB	5.8	9.1	13.9	18.6	23.2	26.9	28.8	29.4	24.2	18.9	13.2	7.5		
		MCWB	4.0	5.9	8.2	11.3	15.5	18.8	19.3	18.7	16.0	13.3	9.9	5.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.7	10.6	12.6	16.0	19.0	22.4	22.7	22.6	19.6	17.5	14.2	11.2		
		MCDB	12.0	16.6	19.9	25.1	26.5	30.1	30.8	31.0	28.4	24.2	19.2	14.0		
	2%	WB	7.1	8.9	10.9	14.2	17.9	21.1	21.5	21.3	18.3	16.1	12.6	8.8		
		MCDB	9.6	13.3	17.4	21.4	24.9	28.8	29.7	29.7	26.4	21.8	16.8	11.3		
	5%	WB	5.6	7.6	9.7	13.0	17.0	20.1	20.6	20.3	17.5	14.9	11.5	7.3		
		MCDB	7.6	11.3	15.6	19.4	23.9	27.4	28.3	28.8	24.7	19.8	14.8	9.3		
	10%	WB	4.1	6.1	8.5	12.0	16.0	19.1	19.8	19.4	16.6	13.8	10.2	5.9		
		MCDB	5.7	8.9	13.4	17.4	22.2	25.9	27.0	27.4	22.8	18.0	13.0	7.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.3	9.0	10.8	11.9	12.5	13.1	13.8	14.5	12.6	11.2	8.9	6.9		
		MCDBR	10.5	13.9	16.1	16.6	16.5	16.8	17.8	18.2	17.1	15.1	12.6	10.3		
		MCWBR	7.3	8.5	8.7	7.9	7.5	7.3	6.9	6.6	7.0	7.8	7.4	7.0		
		MCDBR	9.7	12.7	14.5	14.1	14.8	15.2	15.2	15.5	14.5	13.1	11.1	9.2		
	5% WB	MCWBR	6.8	7.9	8.1	7.1	7.2	7.1	6.8	6.6	6.6	7.3	6.7	6.5		
(40) Clear-Sky Solar Irradiance	taub		0.336	0.369	0.418	0.486	0.470	0.458	0.464	0.480	0.449	0.402	0.364	0.336		
	taud		2.277	2.188	2.113	1.952	2.057	2.137	2.127	2.077	2.128	2.246	2.312	2.314		
	Ebn at Noon		777	809	812	784	810	820	810	781	772	763	739	741		
	Edh at Noon		88	114	139	177	163	151	151	154	136	107	86	78		
(44) All-Sky Solar Radiation	RadAvg		1.52	2.37	3.42	4.43	5.35	6.11	6.29	5.65	4.11	2.82	1.75	1.30		
	RadStd		0.23	0.32	0.42	0.54	0.49	0.53	0.46	0.49	0.34	0.38	0.20	0.19		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.67	N/A	N/A	N/A	N/A	+0.57	N/A	N/A	N/A	N/A
(47) Regional (39 neighbors)		+0.52	N/A	N/A	N/A	+0.37	N/A	N/A	-146	N/A	N/A

Nomenclature: See separate page